A Geometric $L^2(P)$ Version of Sieve-Orthogonalized Empirical Likelihood

Siyan Dong

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Abstract

This note rewrites the sieve-orthogonalized empirical likelihood test in Hilbert-space language. All relevant random variables are assumed to belong to $L^2(P)$, so nuisance fitting, residualization, and score construction can be read as orthogonal projections. The unknown exogenous-control component is approximated by a finite-dimensional sieve subspace, the profiled residual is the orthogonal residual from that subspace, and the empirical likelihood weight is residualized against the same space. The resulting moment is geometrically orthogonal to the fitted nuisance directions. Under the same high-level persistent-predictor regularity conditions and sieve growth window, the empirical likelihood ratio has a standard χ_1^2 limit.

1 Introduction

Predictive regressions often include persistent predictors, such as lagged asset prices or valuation ratios, together with other predetermined state variables. A fixed intercept is too restrictive when those state variables have a nonlinear effect on returns. This paper treats that intercept-like component as an unknown vector in a Hilbert space and tests predictability after projecting it out nonparametrically.

The maintained model is

$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t, \qquad X_t = \theta + \rho_T X_{t-1} + \varepsilon_t, \tag{1}$$

where Y_t is the return, X_{t-1} is a persistent predictor, and $W_{t-1} \in \mathbb{R}^d$ is a stationary exogenous covariate vector. The function $m(\cdot)$ is unrestricted up to sieve smoothness conditions. The null hypothesis is $H_0 : \beta = \beta_0$, with $\beta_0 = 0$ giving the usual test of no mean predictability.

The geometric convention is fixed from the start: each random variable used below is an element of $L^2(P)$, identified up to a.s. equality, with inner product

$$\langle A, B\rangle_P = E_P[AB], \qquad \|A\|_{2,P} = \langle A, A\rangle_P^{1/2}.$$

Thus residuals are not merely algebraic errors. They are orthogonal projection residuals. A nuisance estimate is a projection onto a finite-dimensional sieve space, and a feasible score is useful because it lives asymptotically in the orthogonal complement of that nuisance space.

The main contribution is a sieve-orthogonalized empirical likelihood (EL) construction. The nuisance function m is profiled out under each candidate β. The EL weight is then projected off the same sieve basis. This makes the fitted nuisance component geometrically orthogonal to the estimating equation, while the remaining approximation and projection errors vanish under a transparent growth window for the sieve dimension K.

The framework is motivated by settings such as agricultural commodity returns, where local weather variables may affect supply and returns nonlinearly while lagged financial predictors remain highly persistent. The theory itself is generic: the formal result only requires a stationary exogenous covariate vector satisfying the stated orthogonality and regularity conditions.

2 Model and Sieve-EL Construction

Let

$$\boldsymbol{Y} = (Y_1, \ldots, Y_T)', \qquad \boldsymbol{X}_{lag} = (X_0, \ldots, X_{T-1})'.$$

For the sample problem, use the empirical Hilbert space $\mathbb{R}^T$ with inner product and norm

$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle_T = T^{-1} \boldsymbol{a}' \boldsymbol{b}, \qquad \|\boldsymbol{a}\|_T = \langle \boldsymbol{a}, \boldsymbol{a} \rangle_T^{1/2}.$$

Let $\boldsymbol{P}$ be the $T \times K$ sieve matrix whose t-th row is $\boldsymbol{P}_K(W_{t-1})'$, where

$$\boldsymbol{P}_K(W_{t-1}) = (p_1(W_{t-1}), \ldots, p_K(W_{t-1}))'.$$

When $d > 1$, the basis functions p_j are multivariate. The column space

$$\mathcal{W}_{K,T} = \operatorname{span}\{\boldsymbol{P}_1, \ldots, \boldsymbol{P}_K\} \subset \mathbb{R}^T$$

is the empirical version of the population sieve space generated by functions of W_{t-1}. Define the orthogonal projector and residual-maker

$$\boldsymbol{\Pi}_K = \boldsymbol{P}(\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}', \qquad \boldsymbol{M}_K = \boldsymbol{I}_T - \boldsymbol{\Pi}_K.$$

Here $\boldsymbol{\Pi}_K \boldsymbol{v}$ is the closest point in $\mathcal{W}_{K,T}$ to $\boldsymbol{v}$, and $\boldsymbol{M}_K \boldsymbol{v} \in \mathcal{W}_{K,T}^{\perp_T}$ is its orthogonal residual.

For each candidate value β, the null-imposed profile sieve coefficient and residual are

$$\widehat{\boldsymbol{c}}_K(\beta) = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}), \qquad \widehat{\boldsymbol{u}}(\beta) = \boldsymbol{M}_K(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}). \tag{2}$$

Equivalently,

$$\boldsymbol{P}\widehat{\boldsymbol{c}}_K(\beta) = \operatorname{op}_T(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag} \mid \mathcal{W}_{K,T}), \qquad \widehat{\boldsymbol{u}}(\beta) \perp_T \mathcal{W}_{K,T}.$$

The profiled residual is therefore the empirical distance-minimizing residual after removing the fitted nuisance direction.

Let $w_t = w(X_{t-1})$ be a bounded saturated weight, let $\boldsymbol{w} = (w_1, \ldots, w_T)'$, and define the sieve-orthogonalized weight

$$\boldsymbol{w}^c = \boldsymbol{M}_K \boldsymbol{w}, \qquad w_t^c = (\boldsymbol{w}^c)_t. \tag{3}$$

The key identity is geometric:

$$\boldsymbol{w}^c \in \mathcal{W}_{K,T}^{\perp_T}, \qquad \langle \boldsymbol{p}, \boldsymbol{w}^c \rangle_T = 0 \quad \text{for every } \boldsymbol{p} \in \mathcal{W}_{K,T}.$$

In matrix form this is $\boldsymbol{P}'\boldsymbol{w}^c = \boldsymbol{0}$. Thus $\boldsymbol{w}^c$ is not an ad hoc transformed instrument. It is the raw weight with its nuisance-space component removed. The feasible scalar score is

$$\widehat{Z}_t(\beta) = \widehat{u}_t(\beta) w_t^c. \tag{4}$$

The profile empirical likelihood and log-likelihood ratio are

$$L_T(\beta) = \sup \left\{ \prod_{t=1}^T (T\pi_t) : \pi_t \geq 0, \quad \sum_{t=1}^T \pi_t = 1, \quad \sum_{t=1}^T \pi_t \widehat{Z}_t(\beta) = 0 \right\}, \tag{5}$$

$$\ell_T(\beta) = -2 \log L_T(\beta) = 2 \sum_{t=1}^T \log\{1 + \widehat{\lambda}(\beta) \widehat{Z}_t(\beta)\}, \tag{6}$$

where $\widehat{\lambda}(\beta)$ solves the usual EL multiplier equation.

3 Assumptions and Sieve Growth

Write $\overline{w}_T = T^{-1}\sum_{t=1}^T w_t$, $\widetilde{w}_t = w_t - \overline{w}_T$, and keep the empirical norm $\|\boldsymbol{v}\|_T^2 = T^{-1}\sum_{t=1}^T v_t^2$. All random variables below are in $L^2(P)$. Higher moments are imposed only where they are needed for the oracle EL expansion and maximal bounds.

Assumption 1 (Oracle persistent-predictor score conditions)**.** *The predictor persistence process belongs to a regularity class for which the centered weighted oracle score admits a mixed-normal self-normalized limit. With respect to the enlarged filtration*

$$\mathcal{H}_t = \sigma\{(z_s, \varepsilon_s) : s \le t\} \vee \sigma\{W_s : s \le t\},$$

$E(U_t \mid \mathcal{H}_{t-1}) = 0$, $\sup_t E|U_t|^{2+\delta} < \infty$ *for some* $\delta > 0$, *the raw weight is uniformly bounded, and the conditional-variance and maximum-score conditions needed for self-normalized empirical likelihood hold. In particular, for*

$$A_T = \frac{1}{\sqrt{T}}\sum_{t=1}^T U_t \widetilde{w}_t, \qquad V_T = \frac{1}{T}\sum_{t=1}^T U_t^2 \widetilde{w}_t^2,$$

there exists an a.s. positive, possibly random, variable η^2 *such that*

$$A_T \xrightarrow{\text{st}} MN(0, \eta^2), \qquad V_T \xrightarrow{p} \eta^2, \qquad \max_{t \le T} |U_t \widetilde{w}_t| = o_p(\sqrt{T}). \tag{7}$$

Assumption 2 (Sieve orthogonality and projection)**.** *The first component of* $\boldsymbol{P}_K(W)$ *is one. After centering the remaining basis functions in sample, write the corresponding* $T \times (K-1)$ *matrix as* $\boldsymbol{R}$, *so that* $\mathbf{1}_T'\boldsymbol{R} = \mathbf{0}$ *and* $\operatorname{span}(\boldsymbol{P}) = \operatorname{span}(\mathbf{1}_T, \boldsymbol{R})$. *The primitive population geometry is*

$$\boldsymbol{R}_K(W_{t-1}) \perp_{L^2(P)} \{w(X_{t-1}) - Ew(X_{t-1})\},$$

that is,

$$\mathbb{E}\left[\boldsymbol{R}_K(W_{t-1})\{w(X_{t-1}) - \mathbb{E}w(X_{t-1})\}\right] = \mathbf{0}.$$

The Gram matrix $T^{-1}\boldsymbol{R}'\boldsymbol{R}$ *has eigenvalues bounded away from zero and infinity with probability tending to one. If* $\zeta_K = \max_{t \le T}\|\boldsymbol{R}_t\|$, *then*

$$\left\|T^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}\right\| = O_p(\sqrt{K/T}), \tag{8}$$

$$\left\|T^{-1/2}\boldsymbol{R}'\boldsymbol{U}\right\| = O_p(\sqrt{K}), \tag{9}$$

$$\left\|T^{-1}\boldsymbol{P}'\boldsymbol{U}\right\| = O_p(\sqrt{K/T}). \tag{10}$$

These sample bounds say that the empirical angles between the centered weight, the oracle error, and the growing nuisance space are small enough for projection residuals to approximate their population counterparts.

Assumption 3 (Approximation, rates, and nondegeneracy)**.** *For some* $\boldsymbol{c}_K$,

$$m(W_{t-1}) = \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}_K + r_{K,t},$$

where $\|\boldsymbol{r}_K\|_T = O(a_K)$, $\|\boldsymbol{r}_K\|_\infty = O(a_K)$, *and* $\sqrt{T}a_K \to 0$. *The sieve projector is stable in sup norm on the approximation residual:* $\|\boldsymbol{M}_K\boldsymbol{r}_K\|_\infty = O(a_K)$. *Moreover,*

$$K/\sqrt{T} \to 0, \qquad \zeta_K\sqrt{K/T} \to 0,$$

and $P(\eta^2 \ge \underline{\eta}^2) = 1$ *for some* $\underline{\eta} > 0$. *The EL convex-hull condition holds under the null.*

If m has smoothness s, W is d-dimensional, and $a_K \asymp K^{-s/d}$, then $\sqrt{T} a_K \to 0$ and $K/\sqrt{T} \to 0$ give the rate window

$$T^{d/(2s)} \ll K \ll T^{1/2}, \tag{11}$$

which is nonempty when $s > d$.

4 Main Result

Theorem 1 (Geometric Wilks phenomenon). *Under Assumptions 1–3, under $H_0 : \beta = \beta_0$,*

$$\ell_T(\beta_0) \xrightarrow{d} \chi^2_1.$$

Proof sketch. Assumption 1 supplies the high-level oracle score limit for

$$A_T = T^{-1/2} \sum_t U_t \widetilde{w}_t, \qquad V_T = T^{-1} \sum_t U_t^2 \widetilde{w}_t^2.$$

Lemma 1 shows that projecting onto the growing nuisance space removes only an asymptotically negligible component of the centered weight and oracle error. This is the finite-sample Hilbert-space statement that

$$\boldsymbol{w}^c = \mathrm{op}_T(\boldsymbol{w} \mid \mathcal{W}_{K,T}^{\perp_T}) \approx \widetilde{\boldsymbol{w}}, \qquad \boldsymbol{M}_K \boldsymbol{U} \approx \boldsymbol{U}.$$

Lemma 2 then proves

$$T^{-1/2} \sum_t \widehat{Z}_t(\beta_0) = A_T + o_p(1), \qquad T^{-1} \sum_t \widehat{Z}_t(\beta_0)^2 = V_T + o_p(1).$$

The denominator equivalence is a distance statement: the feasible score vector and the oracle score vector have empirical distance $o_p(1)$, so Cauchy–Schwarz gives equality of their squared lengths up to $o_p(1)$. The standard scalar EL expansion then yields $\ell_T(\beta_0) = S_T^2/\widehat{V}_T + o_p(1)$, and the result follows by Slutsky's theorem. Full details are in Appendix A.

Corollary 1 (No-predictability test). *For $H_0 : \beta = 0$, reject at asymptotic level α whenever*

$$\ell_T(0) > \chi^2_{1,1-\alpha}.$$

A confidence set for β is $\{\beta : \ell_T(\beta) \le \chi^2_{1,1-\alpha}\}$.

5 Interpretation: Weather Shocks and Commodity Returns

The formal model treats W_{t-1} as a generic stationary exogenous covariate vector. A motivating application is commodity-return prediction with local weather conditions. For cocoa futures, for example, precipitation, temperature, humidity, or other regional weather variables can affect expected supply and returns through a nonlinear response surface $m(W_{t-1})$. At the same time, lagged prices and related financial predictors may be highly persistent.

The geometric interpretation separates these roles. The sieve profile projects returns onto the weather-response space. The orthogonalized weight projects the raw predictor weight onto the orthogonal complement of that same space. The empirical likelihood score therefore tests the incremental predictive content of X_{t-1} using only the component that is geometrically separated from the nuisance directions.

6 Conclusion

The method extends fixed-intercept predictive-regression EL to settings with an unknown exogenous-control function. In the $L^2(P)$ version, the proof isolates the new sieve step by reading profiling and residualization as orthogonal projections. Under the rate window $T^{d/(2s)} \ll K \ll T^{1/2}$, the projection errors are small enough that the sieve-profiled and sieve-orthogonalized feasible score is asymptotically equivalent to the centered oracle score, and the empirical likelihood ratio retains the standard χ_1^2 limit for mean predictability.

A Proofs

Throughout the appendix, all notation is as defined in the main text. Under $H_0 : \beta = \beta_0$, write

$$\boldsymbol{Y} - \beta_0 \boldsymbol{X}_{lag} = \boldsymbol{P}\boldsymbol{c}_K + \boldsymbol{r}_K + \boldsymbol{U}, \qquad \widehat{\boldsymbol{u}}(\beta_0) = \boldsymbol{M}_K(\boldsymbol{U} + \boldsymbol{r}_K).$$

The equalities involving $\boldsymbol{M}_K$ are projection equalities in the empirical Hilbert space $(\mathbb{R}^T, \langle\cdot,\cdot\rangle_T)$.

Lemma 1 (Negligibility of the growing sieve projection)**.** *Under Assumptions 2–3,*

$$\|\boldsymbol{w}^c - \widetilde{\boldsymbol{w}}\|_T = O_p(\sqrt{K/T}), \tag{12}$$

$$\max_{t \le T} |w_t^c - \widetilde{w}_t| = O_p(\zeta_K \sqrt{K/T}) = o_p(1), \tag{13}$$

$$T^{-1/2} \sum_{t=1}^{T} U_t (w_t^c - \widetilde{w}_t) = O_p(K/\sqrt{T}) = o_p(1), \tag{14}$$

$$\|\boldsymbol{\Pi}_K \boldsymbol{U}\|_T = O_p(\sqrt{K/T}), \tag{15}$$

$$\max_{t \le T} |(\boldsymbol{\Pi}_K \boldsymbol{U})_t| = O_p(\zeta_K \sqrt{K/T}) = o_p(1). \tag{16}$$

Proof. Because the sieve contains a constant and $\boldsymbol{1}_T'\boldsymbol{R} = \boldsymbol{0}$, the projection of $\boldsymbol{w}$ on $\operatorname{span}(\boldsymbol{1}_T, \boldsymbol{R})$ first removes the constant component $\overline{w}_T \boldsymbol{1}_T$ and then projects $\widetilde{\boldsymbol{w}}$ on $\operatorname{span}(\boldsymbol{R})$. Hence

$$\boldsymbol{w}^c = \widetilde{\boldsymbol{w}} - \boldsymbol{R}\widehat{\boldsymbol{\delta}}_K, \qquad \widehat{\boldsymbol{\delta}}_K = (\boldsymbol{R}'\boldsymbol{R})^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}.$$

The vector $\boldsymbol{R}\widehat{\boldsymbol{\delta}}_K$ is the empirical orthogonal projection of $\widetilde{\boldsymbol{w}}$ onto $\operatorname{span}(\boldsymbol{R})$. The Gram condition and (8) imply

$$\|\widehat{\boldsymbol{\delta}}_K\| = O_p(\sqrt{K/T}).$$

Therefore its squared projection length is

$$\|\boldsymbol{R}\widehat{\boldsymbol{\delta}}_K\|_T^2 = \widehat{\boldsymbol{\delta}}_K'(T^{-1}\boldsymbol{R}'\boldsymbol{R})\widehat{\boldsymbol{\delta}}_K = O_p(K/T),$$

which proves (12). The maximum bound follows from the row-length inequality

$$\max_t |\boldsymbol{R}_t'\widehat{\boldsymbol{\delta}}_K| \le \zeta_K \|\widehat{\boldsymbol{\delta}}_K\| = O_p(\zeta_K \sqrt{K/T}).$$

The score difference is the inner product between $\boldsymbol{U}$ and the small projected component:

$$T^{-1/2}\boldsymbol{U}'(\boldsymbol{w}^c - \widetilde{\boldsymbol{w}}) = -(T^{-1/2}\boldsymbol{R}'\boldsymbol{U})'\widehat{\boldsymbol{\delta}}_K = O_p(\sqrt{K})O_p(\sqrt{K/T}) = O_p(K/\sqrt{T}).$$

Finally, with $\widehat{\boldsymbol{\gamma}}_U = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'\boldsymbol{U}$, (10) implies $\|\widehat{\boldsymbol{\gamma}}_U\| = O_p(\sqrt{K/T})$. Applying the same projection-length identity and the same row-length inequality to $\boldsymbol{\Pi}_K\boldsymbol{U} = \boldsymbol{P}\widehat{\boldsymbol{\gamma}}_U$ yields (15) and (16). $\square$

Lemma 2 (Feasible-oracle score and variance equivalence)**.** *Under Assumptions 1–3, under* $H_0 : \beta = \beta_0$*,*

$$S_T := T^{-1/2}\sum_{t=1}^{T}\widehat{Z}_t(\beta_0) = T^{-1/2}\sum_{t=1}^{T}U_t\widetilde{w}_t + o_p(1), \tag{17}$$

$$\widehat{V}_T := T^{-1}\sum_{t=1}^{T}\widehat{Z}_t(\beta_0)^2 = T^{-1}\sum_{t=1}^{T}U_t^2\widetilde{w}_t^2 + o_p(1), \tag{18}$$

$$\max_{t\le T}|\widehat{Z}_t(\beta_0)| = o_p(\sqrt{T}). \tag{19}$$

Consequently, $\widehat{V}_T \xrightarrow{p} \eta^2$*.*

Proof. Because $\boldsymbol{M}_K$ is an orthogonal projector and $\boldsymbol{w}^c = \boldsymbol{M}_K\boldsymbol{w}$,

$$S_T = T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_K)'\boldsymbol{w}^c.$$

Lemma 1 gives $T^{-1/2}\boldsymbol{U}'\boldsymbol{w}^c = T^{-1/2}\boldsymbol{U}'\widetilde{\boldsymbol{w}} + o_p(1)$. For the approximation residual, Cauchy–Schwarz in $(\mathbb{R}^T, \langle\cdot,\cdot\rangle_T)$ gives

$$|T^{-1/2}\boldsymbol{r}_K'\boldsymbol{w}^c| \le \sqrt{T}\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}^c\|_T \le \sqrt{T}\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}\|_T = o_p(1),$$

where the second inequality uses the contraction property of orthogonal projection, $\|\boldsymbol{M}_K\boldsymbol{w}\|_T \le \|\boldsymbol{w}\|_T$. This proves (17).

For the denominator, define the feasible-oracle distance components

$$\Delta_{u,t} = \widehat{u}_t(\beta_0) - U_t, \qquad \Delta_{w,t} = w_t^c - \widetilde{w}_t, \qquad \Delta_{Z,t} = \widehat{Z}_t(\beta_0) - U_t\widetilde{w}_t.$$

Then $\boldsymbol{\Delta}_u = -\boldsymbol{\Pi}_K\boldsymbol{U} + \boldsymbol{M}_K\boldsymbol{r}_K$, so by the projection lemma and approximation stability,

$$\|\boldsymbol{\Delta}_u\|_T = O_p(\sqrt{K/T} + a_K) = o_p(1).$$

Moreover, $\max_t|\Delta_{w,t}| = o_p(1)$, $\max_t|w_t^c| = O_p(1)$, and $T^{-1}\sum_t U_t^2 = O_p(1)$. Since

$$\Delta_{Z,t} = \Delta_{u,t}w_t^c + U_t\Delta_{w,t},$$

we have

$$\|\boldsymbol{\Delta}_Z\|_T^2 \le 2\max_t|w_t^c|^2\|\boldsymbol{\Delta}_u\|_T^2 + 2\max_t|\Delta_{w,t}|^2T^{-1}\sum_{t=1}^{T}U_t^2 = o_p(1).$$

Thus the feasible score vector $\widehat{\boldsymbol{Z}}(\beta_0)$ and the oracle score vector $(U_t\widetilde{w}_t)_{t=1}^T$ are $o_p(1)$ apart in empirical distance. If $V_T = T^{-1}\sum_t U_t^2\widetilde{w}_t^2$, then

$$\widehat{V}_T - V_T = \frac{2}{T}\sum_{t=1}^{T}U_t\widetilde{w}_t\Delta_{Z,t} + \|\boldsymbol{\Delta}_Z\|_T^2.$$

Another Cauchy–Schwarz application gives

$$|\widehat{V}_T - V_T| \le 2V_T^{1/2}\|\boldsymbol{\Delta}_Z\|_T + \|\boldsymbol{\Delta}_Z\|_T^2 = o_p(1),$$

because $V_T = O_p(1)$ by Assumption 1. This proves (18), and $\widehat{V}_T \to_p \eta^2$ follows from (7).

Finally, $\max_t|U_t| = o_p(\sqrt{T})$ follows from the uniform $(2+\delta)$-moment bound. Together with (16), the sup-norm stability of $\boldsymbol{M}_K\boldsymbol{r}_K$, and $\max_t|w_t^c| = O_p(1)$, this proves (19). □

Proof of Theorem 1. By Assumption 1 and Lemma 2,

$$S_T \xrightarrow{\text{st}} MN(0, \eta^2), \qquad \widehat{V}_T \xrightarrow{p} \eta^2.$$

Hence $S_T^2/\widehat{V}_T \xrightarrow{d} \chi_1^2$. The convex-hull condition, $\widehat{V}_T \to_p \eta^2 > 0$, and $\max_t |\widehat{Z}_t(\beta_0)| = o_p(\sqrt{T})$ imply the standard scalar-EL bounds $\widehat{\lambda} = O_p(T^{-1/2})$ and $\max_t |\widehat{\lambda}\widehat{Z}_t| = o_p(1)$. Expanding the multiplier equation gives

$$\widehat{\lambda} = \frac{\sum_{t=1}^T \widehat{Z}_t(\beta_0)}{\sum_{t=1}^T \widehat{Z}_t(\beta_0)^2} + o_p(T^{-1/2}).$$

A second-order expansion of (6) yields

$$\ell_T(\beta_0) = \frac{\{T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0)\}^2}{T^{-1} \sum_{t=1}^T \widehat{Z}_t(\beta_0)^2} + o_p(1) = \frac{S_T^2}{\widehat{V}_T} + o_p(1).$$

The result follows by Slutsky's theorem. □